

Complex Computing Problems and Projects

CS406-Digital Image Processing

Summer 2026

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General Information

Course Learning Outcomes (CLOs)

- CLO-1: Describe Fundamentals of Digital Image Processing
- CLO-2: Apply Image Enhancement and Filtering Techniques
- CLO-3: Analyze Image Segmentation and Morphological Processing
- CLO-4: Design Image Processing Algorithms

Assessment Rubric

- Problem analysis and constraints (3 marks)
- DIP Design Pipeline and flowcharts (5 marks)
- Interdependence of modules (2 marks)
- Report originality and flow (2 marks)
- Viva (3 marks)

Submission Requirements

- LaTeX-generated PDF (10 pages, double column)
- IEEE format references
- Overleaf editable link
- YouTube video presentation
- Plagiarism below 15%
- Five(5) member Team with photos and biographies

Important Dates

- Project Selection (Get Instructor Approval): Week 2
- Submission deadline: Monday, Week 8
- Maximum marks: 15/100

Deliverables Timeline

- Weeks 2-3: Initial Proposal
- Weeks 5: Mid-Semester Progress Report
- Week 7: Final System
- Weeks 8: Final Report and Presentation

Option-1 Select Project 1: Smart City Surveillance System

Problem Statement

The city of Islamabad has implemented a smart surveillance system for real-time monitoring of public spaces. However, the system struggles with poor image quality, noisy environments, and varying lighting conditions, making it challenging to identify vehicles and pedestrians effectively.

Key Requirements

- Real-Time Performance vs. Processing Quality
- Resource Constraints vs. Performance
- Noise Removal vs. Detail Preservation
- Compression vs. Image Fidelity
- Security and Privacy vs. System Accessibility

Technical Tasks

1. Spatial Filtering (Mean, Median, Laplacian)
2. Frequency Domain Filtering (Fourier transforms)
3. Image Restoration and Reconstruction
4. Image Segmentation (Edge-based and Region-based)
5. Image Compression (JPEG, Wavelets)
6. Object Recognition (Optional: YOLO, Faster R-CNN)

Assessment Metrics

- PSNR, SSIM for image quality
- Processing speed for real-time performance
- IoU, Dice coefficient for segmentation
- Precision, Recall, F1-score for recognition

Deliverables

- LaTeX report (10 pages, IEEE format)
- Complete implementation code
- Performance analysis
- Video presentation
- Overleaf editable link

Option-2 Select Project 2: Satellite Image Analysis for Natural Disaster Prediction

Problem Statement

Design a satellite image analysis system capable of predicting natural disasters (floods, wildfires) and assessing post-disaster damage by analyzing before-and-after satellite images under various constraints.

Key Requirements

- Time-sensitive predictions vs. Data accuracy
- Handling diverse disasters with single system
- Limited computational power
- Geospatial accuracy vs. Large-scale coverage

Technical Tasks

1. Time-series satellite image analysis
2. Change detection algorithms
3. Noise reduction and image enhancement
4. Disaster prediction modeling
5. Damage assessment techniques

Suggested Tools

- Google Earth Engine, ArcGIS
- Open-source GIS libraries
- Python with OpenCV, scikit-image
- Change detection algorithms

Option-3 Select Project 3: Autonomous Crop Monitoring System for Precision Agriculture

Problem Statement

Design and implement an autonomous image analysis system using drone-captured images to monitor crop health, detect diseases, and provide actionable insights to farmers under resource constraints.

Key Requirements

- Limited computational resources vs. Large-scale processing
- Scalability vs. Real-time monitoring
- Environmental variability vs. Detection accuracy
- Cost constraints vs. Advanced capability

Technical Tasks

1. Crop health monitoring using vegetation indices (NDVI)
2. Disease and stress detection
3. Crop classification and coverage estimation
4. Environmental adaptation techniques
5. Resource optimization for edge devices

Suggested Techniques

- Vegetation indices (NDVI, SAVI)
- Machine learning classification
- Image segmentation for crop detection
- Noise reduction for environmental variations
- Edge computing optimization

Learning Objectives

- Advanced image processing techniques
- Scalability and resource management
- Problem solving under constraints
- Real-world agricultural applications

Dataset Resources

For Project 1: Smart City Surveillance

- **Custom Face Mask Dataset:** A dataset of 14,535 images for face mask detection, created for a real-time surveillance case study [1]. Kaggle Link
- **Cityscapes Dataset:** A large-scale dataset for semantic urban scene understanding, containing a diverse set of stereo video sequences recorded in streets from 50 different cities [2].
- **KITTI Dataset:** A popular dataset for autonomous driving research, providing rich urban environment data [3].
- **CAVIAR Dataset:** A dataset of video sequences for surveillance and human behavior analysis [4].
- **Lahore Urban Traffic Observations Dataset (2025):** Traffic flow data from major corridors in Lahore, Pakistan, collected via CCTV and sensors [5]. Kaggle Link

For Project 2: Satellite Image Analysis for Disaster Prediction

- **NASA's Harmonized Landsat and Sentinel-2 (HLS) Dataset:** A near-global, 30m resolution dataset with high temporal frequency (2-3 days), available on Google Earth Engine. This is ideal for change detection and disaster monitoring [6].
- **FloodNet Dataset:** A benchmark dataset of high-resolution UAV images captured post-Hurricane Harvey, containing diverse classes like flooded/non-flooded roads and buildings [7].
- **Global Natural Disasters (2000–2025):** A curated collection of natural events from NASA's EONET, including wildfires, floods, and storms with geospatial data [8]. Kaggle Link
- **UC Merced Land Use Dataset:** A 21-class land use remote sensing image dataset, often used for benchmarking classification and enhancement techniques [9].
- **Satellite Imagery from Kaggle:** A dataset containing images of various terrains (mountains, beaches, ice, rivers) for evaluating image processing models [10].

For Project 3: Autonomous Crop Monitoring

- **Smart Pest Detection Agriculture Dataset:** An IoT and image-based dataset for pest detection in agriculture, containing environmental and image-derived features [11]. Kaggle Link
- **UAV-Based Agricultural Image Dataset:** A collection of UAV images captured over agricultural fields treated with different soil amendments (e.g., gypsum, charcoal) [12]. Kaggle Link

- **Crop-Weed Segmentation UAV RGB+Indices:** High-resolution UAV images of sorghum fields with segmentation masks for crops, weeds, and soil. Includes vegetation indices [13]. Kaggle Link
- **NeuroRF FarmSense Dataset:** A crop recommendation dataset from Kaggle containing 2,200 entries and seven critical attributes for crop selection [14].

References

- [1] S. Burhan, "Face Mask Dataset," Kaggle, 2023. [Online]. Available: <https://www.kaggle.com/datasets/shiekhburhan/face-mask-dataset>
- [2] M. Cordts et al., "The Cityscapes Dataset for Semantic Urban Scene Understanding," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 3213-3223.
- [3] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for Autonomous Driving? The KITTI Vision Benchmark Suite," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2012, pp. 3354-3361.
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- [5] E. M. Khan, "Lahore Urban Traffic Observations Dataset 2025," Kaggle, 2025. [Online]. Available: <https://www.kaggle.com/datasets/emmmmj/lahore-urban-traffic-observations-dataset-2025>
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- [7] M. Rahnemoonfar et al., "FloodNet: A High Resolution Aerial Imagery Dataset for Post Flood Scene Understanding," *IEEE Access*, vol. 9, pp. 118744-118756, 2021.
- [8] S. Bhattarai, "Global Natural Disasters (2000–2025)," Kaggle, 2025. [Online]. Available: <https://www.kaggle.com/datasets/sujanbhattarai108/global-natural-disasters-20002025>
- [9] Y. Yang and S. Newsam, "Bag-of-Visual-Words and Spatial Extensions for Land-Use Classification," in *Proc. ACM SIGSPATIAL Int. Conf. Adv. Geogr. Inf. Syst.*, 2010, pp. 270-279.
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